



ENTERPRISE MILESTONE SPECIFICATION

Module 9: Executive Management Dashboard, 8-Module BI & Mobile Cockpit

Consolidated analytical data warehouse, cross-module KPI calculation engines, interactive management dashboards, and executive mobile cockpit for Popular Auto Spare & A/C Parts Co. W.L.L.

MODULE IDENTIFIER	BA REFERENCE DOCUMENT	TOTAL ENGINEERING EFFORT
SL-POP-ERP-MS-009	DOC-009 v1.0 (04-July-2026)	4 Working Weeks (1.0 Month)
FIXED PROJECT COST	TARGET ARCHITECTURE	DOCUMENT STATUS
BD 1,523.636	Real-Time Data Warehouse, OLAP Data Marts & Mobile BI	FINAL PROPOSAL

Section 1.0

Executive Summary & Master ERP Context

Strategic Alignment

The **Executive Management KPI & BI Dashboard Platform** (Module 9) extracts operational feeds from all 8 upstream ERP modules (Item Master, Procurement, Inventory, Sales, Finance, Admin, HRMS, and Hardware) into an aggregated real-time data warehouse. It empowers Board of Directors and Executive Leadership of Popular Auto Spare & A/C Parts Co. W.L.L. with instantaneous visibility into gross margins, revenue velocity, branch sales leaderboards, dead stock exposure, cash flow liquidity, and operational bottlenecks.

Core Objective & Business Purpose

To establish an automated, low-latency Business Intelligence and executive analytics platform consolidating star-schema OLAP data marts, multi-branch comparative sales leaderboards, gross profit velocity trackers, dead stock diagnostic filters, and mobile-responsive management cockpits.

MASTER ERP TRANSFORMATION ARCHITECTURE (9 CORE PROCESS MODULES)

CURRENT SCOPE: MODULE 09 ACTIVE

- 01 PCode Gen & Item Master
- 02 Vendor & Purchase Flow
- 03 Stock Verification & Location
- 04 Sales & POS Checkout
- 05 Finance, Tax & VAT Accounting
- 06 Administration & Security
- 07 HRMS & Biometric Payroll
- 08 Hardware & Barcode Infrastructure
- 09 Executive BI & Mobile Analytics



Cross-Module OLAP Warehouse

Consolidated analytical warehouse aggregating real-time data feeds across all 8 ERP operational modules.

- OLAP Star Schema
- ETL Pipeline
- Data Marts



Executive Dashboard & Visual KPI

Real-time revenue, gross margin, branch sales leaderboards, AR aging, and inventory turn charts.

- Visual Charts
- Profit Velocity
- Branch Leaderboard



Mobile Cockpit & Early Warnings

Responsive smartphone executive cockpit with automated low-margin and dead stock threshold alerts.

- Mobile Cockpit
- Early Warnings
- RBAC Security

Section
2.0

Module 9: Functional Scope & Architecture
Breakdown (BA Ref: DOC-009)

Full Specification
Mapping

Derived directly from **DOC-009: Executive Management KPIs & BI Analytics Business Analysis Report**, the implementation scope is structured across 6 core analytical pillars:

2.1 Cross-Module Data Warehouse & Real-Time ETL Pipeline

- **Star-Schema Data Marts:** Optimized fact and dimension tables consolidating transactions from Sales, Inventory, Purchasing, and GL.
- **Sub-Minute Incremental ETL:** Micro-batch sync daemon aggregating transactional metrics into memory-cached analytical summaries.
- **Cross-Branch Data Normalization:** Unified currency, calendar, and brand categorization bridging multi-outlet performance metrics.

2.2 Sales, Margin & Revenue Performance Intelligence

- **Real-Time Sales Cockpit:** Hourly and daily gross revenue tracking against monthly sales targets and historical run-rates.
- **Gross Margin Analytics:** Item-level, category-level, and brand-level margin analysis highlighting erosion and price discounts.
- **Branch & Counter Comparison:** Interactive leaderboard comparing branch basket sizes, footfall conversion, and salesperson performance.

2.3 Procurement, Supplier Performance & Spend Analytics

- **Supplier Lead Time Tracking:** Comparison of promised vs. actual shipment arrival dates and vendor fulfillment fill-rates.
- **Spend Concentration Analysis:** Top 20 supplier volume matrices and purchase price variance (PPV) trend tracking.
- **Landed Cost Analytics:** Comparison of freight, customs duty, and inland transport costs across overseas supplier shipments.

2.4 Inventory Turn, Slow-Moving & Dead Stock Diagnostics

- **Stock Aging Matrix:** 0–30, 31–60, 61–90, 91–180, and 180+ day inventory valuation breakdowns across all branches.
- **Dead Stock Warning Radar:** Early warning alarms for spare parts with zero sales movement exceeding 90 consecutive days.
- **Inventory Turnover Ratio (ITR):** Real-time calculation of stock turn velocity per car make, model year, and product category.

2.5 Financial Liquidity, Working Capital & Cash Flow Cockpit

- **Accounts Receivable (AR) Health:** Aging analysis of workshop credit balances, DSO (Days Sales Outstanding), and overdue alerts.
- **Cash Flow Forecast Engine:** 30/60/90-day liquidity projection combining upcoming vendor AP bills and expected customer collections.
- **VAT & Tax Liability Tracker:** Real-time estimation of quarterly Bahrain NBR 10% VAT liability and input tax recovery balances.

2.6 Mobile Executive Dashboard & Automated Anomaly Alerts

- **Smartphone Executive Cockpit:** Responsive progressive web app for CEO/BOD featuring high-level executive cards and KPI dials.
- **Automated Anomaly Push Alerts:** Proactive SMS/Email/Portal notifications for unusual margin drops, abnormal discounts, or stock adjustments.
- **Role-Based Access Governance:** Granular multi-tiered executive security restricting sensitive financial figures to authorized directors.

Section 3.0 Dedicated BI Engineering Resource Allocation (4 Weeks / 1.0 Month)

Standard Rate Card

In strict accordance with the **SaNDS Lab Dedicated Engineering Rate Card**, the resource allocation and cost breakdown for the 4-week (1.0 month / 20 working days) delivery of Module 9 is structured as follows (all figures in Bahraini Dinars – BHD):

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	1.0 Month (4 Wks)	BD 545.455
Lead BI & OLAP Data Warehouse Architect	BD 318.182	BD 79.546	BD 1.989	1.0 Month (4 Wks)	BD 318.182
Lead Front-End & Visualization Engineer (React/Charts)	BD 227.273	BD 56.818	BD 1.420	1.0 Month (4 Wks)	BD 227.273
Back-End API & ETL Pipeline Engineer	BD 227.273	BD 56.818	BD 1.420	1.0 Month (4 Wks)	BD 227.273
QA, Security & Mobile Optimization Engineer	BD 204.545	BD 51.136	BD 1.278	1.0 Month (4 Wks)	BD 204.545
Total Dedicated Engineering Team	BD 1,523.636 / mo	BD 380.909 / wk	BD 9.518 / hr	4 Weeks (1.0 Mo)	BD 1,523.636

**Fixed Price Guarantee**

Total project investment for Module 9 is fixed at **BD 1,523.636**. No hidden deployment, integration, or hourly surcharges apply within the agreed functional scope.

Section
4.0

Detailed Milestone Breakdown & Deliverables (4-Week Roadmap)

Sprint Execution
Plan

The implementation is organized into 4 sequential 1-week milestones (25.00% each), guaranteeing agile delivery, continuous validation, and transparent deliverable acceptance:

1

Milestone 9.1: OLAP Star-Schema Data Marts, Aggregation Engine & ETL Pipelines

Week 1 (1 Wk)

Design and deployment of analytical star-schema data warehouse, dimensional modeling, automated ETL extraction daemons across Modules 1 through 8, and caching layer configuration.

- ✓ **Data Warehouse Schema:** Star-schema facts/dimensions for sales, stock, purchasing, and GL accounts.

✓ **Aggregation Caching:** Pre-calculated hourly and daily summary caches for instantaneous query responses.
- ✓ **ETL Sync Daemons:** Sub-minute micro-batch synchronization pipelines pulling live transactional records.

2

Milestone 9.2: Real-Time Executive KPI Calculation Engines & Anomaly Alert Triggers

Week 2 (1 Wk)

Engineering of gross margin algorithms, branch sales ranking engine, dead stock aging matrices, AR collection health metrics, and automated anomaly alert notification triggers.

- ✓ **KPI Calculation Engines:** Real-time formulas for gross margin %, DSO, stock turn ratio, and sales targets.

✓ **Anomaly Alert Rules:** Threshold breach alarms for low-margin transactions and credit limit overrides.
- ✓ **Dead Stock Aging Radar:** Multi-tier stock aging classification and zero-movement warning triggers.

3

Milestone 9.3: Interactive BI Dashboard Web UI, Data Visualizations & Drill-Down Grids

Week 3 (1 Wk)

Development of rich interactive web UI featuring revenue charts, branch leaderboards, margin heatmaps, multi-dimensional filtering, and transaction drill-down capabilities.

- ✓ **Interactive Visual Charts:** Revenue velocity line charts, category pie breakdowns, and branch bar charts.
- ✓ **Multi-Dimensional Filters:** Dynamic slicing by branch, date range, product family, brand, and salesman.
- ✓ **Drill-Down Data Grids:** One-click navigation from top-level KPI down to source transactional invoice lines.

4

Milestone 9.4: Mobile Executive Cockpit, RBAC Authorization & Production Go-Live

Week 4 (1 Wk)

Deployment of mobile-optimized executive cockpit for smartphones/tablets, role-based access security controls, executive UAT verification, and final production sign-off.

- ✓ **Mobile Executive Cockpit:** Responsive progressive web interface optimized for executive mobile access.
- ✓ **Executive RBAC Security:** Strict multi-level authorization locking confidential margin and profit figures.
- ✓ **Production Acceptance:** Comprehensive validation by Board of Directors & executive management sign-off.

Section 5.0

Milestone Payment Structure & Invoicing Schedule (4-Week Model)

Commercial Billing Governance

Billing is structured across 4 sequential verification gates tied directly to verifiable deliverable sign-offs:

GATE	MILESTONE DELIVERABLE FOCUS	TIMELINE	% FEE	INVOICE AMOUNT (BD)	VERIFICATION CRITERIA
Gate 9.1	OLAP Schema, Data Warehouse & ETL Pipelines	End of Wk 1	25.00%	BD 380.909	Cross-module ETL extraction validated against live DB
Gate 9.2	Real-Time KPI Calculation Engines & Anomaly Alerts	End of Wk 2	25.00%	BD 380.909	POS, dead stock, and AR formulas verified by Finance
Gate 9.3	Visual Dashboard UI & Drill-Down Analytics	End of Wk 3	25.00%	BD 380.909	Interactive web UI charts & drill-downs verified
Gate 9.4	Mobile Executive Cockpit, RBAC & Final Sign-Off	End of Wk 4	25.00%	BD 380.909	Mobile cockpit approved & final executive acceptance
Total Module 9 Investment (Fixed Deliverable Contract)			100.00%	BD 1,523.636	Complete Production Delivery

Section

6.0

Working Calendar, SLA, 15-Day Grace Period

& Legal Terms

Master Service Level Agreement

This milestone schedule is governed strictly under the overarching Master Payment Governance & Service Level Framework (SL-POP-ERP-PAY-001):

 Standard Working Calendar & Effort Accounting

All engineering effort calculations are structured upon a standard 5-day working week (Sunday through Thursday), 8 working hours per day (40 hours/week, 160 hours/month). Kingdom of Bahrain statutory public holidays are observed.

 15-Day Verification Window & Deemed Acceptance

Upon milestone deployment notification, the Client is granted a 15-calendar-day UAT verification period. If no material defect is logged in writing within 15 days, milestone acceptance and invoice authorization occur automatically.

 90-Day Production Bug-Fix Warranty

A comprehensive 90-calendar-day post-go-live warranty covers all functional discrepancies, calculation anomalies, and system defects originating from the agreed DOC-009 specification without additional charge.

 2-Year Resource Rate Protection Guarantee

Engineering resource rates defined in Section 3.0 remain legally fixed for twenty-four (24) months from contract execution, ensuring predictable costs for any post-implementation change requests.

Section

7.0

Stakeholder Authorization & Digital Sign-Off

Console

Cryptographic Audit Trail

By executing digital signatures below, all authorized representatives formally validate the functional architecture, 4-week delivery roadmap, resource allocation pricing, and invoicing governance defined in this document:

FIRST PARTY – CLIENT / END-USER

Popular Auto Spare & A/C Parts Co. W.L.L.

Awaiting Client Signature

Signatory: Authorized Executive Director

Status: Pending Digital Execution

SECOND PARTY – LEAD IMPLEMENTATION PARTNER

SaNDS Lab Software Solutions

Awaiting SaNDS Lab Signature

Signatory: Lead Solutions Architect

Status: Pending Digital Execution

THIRD PARTY – ADVISORY & ASSURANCE PARTNER

UniGlobal Business Solutions

Awaiting UniGlobal Signature

Signatory: Principal Consultant

Status: Pending Digital Execution